

ENGINEERS WITHOUT BORDERS - SVCE STUDENT CHAPTER FIELD VISIT SPACE ZONE INDIA

DATE: 19/03/2026

VENUE: Space Zone India, Thaiyur, Chennai - 603103,



TABLE OF CONTENTS

S/NO	TITLE	PG/NO
1	EXECUTIVE SUMMARY	3
2	OBJECTIVES	4
3	IMPACT ASSESSMENT	4
4	CHRONOLOGICAL PROCEEDINGS OF THE VISIT	5
5	CONCLUSION	7
6	ANNEXURE – I (ATTENDEES LIST)	8
7	ANNEXURE - II (GALLERY)	9
8	ANNEXURE – III (FEEDBACK FORM)	12

EXECUTIVE SUMMARY

The field visit to Space Zone India, organized by the Engineers Without Borders (EWB) - SVCE Student Chapter, was conducted on **19 April 2026** as an immersive experiential learning program aimed at providing students with direct exposure to advanced space technologies and their real-world engineering applications. The initiative was designed to bridge the gap between theoretical academic knowledge and practical implementation in the domains of aerospace engineering, satellite systems, and propulsion technologies.

The program commenced with a series of technical sessions focusing on rocket technology and propulsion system analysis. Participants were provided with detailed insights into solid, liquid, and hybrid propulsion mechanisms, with emphasis on combustion dynamics, thrust generation, and efficiency considerations. The discussion further extended to research and development methodologies, highlighting experimental validation techniques such as combustion testing and variation analysis used to optimize propulsion system performance. These sessions enabled students to understand the iterative, analytical, and data-driven nature of aerospace engineering design.

Participants were also introduced to Space Zone India's ongoing projects, particularly the RHUMI series of rockets. The **RHUMI-1 (2024)** model, featuring movable fins and compact satellite integration, was discussed as a practical example of design innovation and system optimization. Additionally, updates regarding the **RHUMI Twin system** provided insights into next-generation advancements, reflecting the organization's commitment to developing scalable and efficient space technology solutions.

A key highlight of the visit was the interactive session with the **CEO and Founder of Space Zone India**, who shared valuable insights on the evolving landscape of the space industry, emerging technological trends, and the importance of innovation-driven engineering. During this session, the **official inauguration of the EWB SVCE student-developed website** was also conducted. The launch symbolized the integration of digital and engineering capabilities, and the initiative was acknowledged and appreciated for its relevance and execution.

The latter half of the program focused on laboratory-based learning, providing hands-on exposure to various critical engineering domains. The **aerodynamics laboratory session** demonstrated airflow behavior, lift and drag forces, and their applications in aircraft, rockets, and drone systems. The **rocket integration laboratory** provided a detailed understanding of rocket structural components, material selection, and internal configuration elements such as fuel storage systems. Furthermore, the **satellite laboratory** introduced participants to different classifications of miniaturized satellites, including Pico, Femto, Cube, and Can satellites, along with their embedded systems, power electronics, and PCB-based subsystem integration.

In addition, the sessions covered broader aspects of satellite technology, including applications in communication, environmental monitoring, and navigation systems. Concepts such as the

Doppler effect in satellite communication were also discussed, highlighting the interdisciplinary nature of space technology that integrates principles from mechanical, electrical, and computational engineering.

Overall, the field visit successfully achieved its objectives by delivering a comprehensive learning experience that combined theoretical understanding, practical exposure, and industry interaction. It enhanced the technical competence of the participants, encouraged innovation and critical thinking, and aligned with the vision of Engineers Without Borders in leveraging engineering solutions for societal and technological advancement.

OBJECTIVES

The primary objectives of the visit to Space Zone India were:

- To provide experiential learning in space technology, including rocket propulsion systems and satellite engineering.
- To develop a fundamental understanding of aerospace concepts such as aerodynamics, combustion dynamics, and system integration.
- To bridge theoretical knowledge with real-world applications through laboratory exposure and expert-led demonstrations.
- To gain insights into different propulsion mechanisms, including solid, liquid, and hybrid systems, and their operational significance.
- To understand satellite classifications (Pico, Femto, Cube, etc.) and their embedded systems, including power electronics and PCB design.
- To foster innovation and technical curiosity through interaction with industry professionals and exposure to ongoing projects such as the RHUMI series.
- To encourage interdisciplinary learning by integrating concepts from mechanical, electronics, and computational engineering domains.

IMPACT ASSESSMENT

The field visit to Space Zone India created a strong academic and technical impact by improving participants' understanding of space technologies and their real-world engineering applications. The laboratory sessions strengthened practical knowledge through exposure to aerodynamics, rocket integration, and satellite subsystem development. The visit also aligned with the **United Nations Sustainable Development Goals, particularly SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure)** by promoting experiential learning and technological advancement.

ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS



CHRONOLOGICAL PROCEEDINGS OF THE VISIT

SESSION I: TECHNICAL 09:50 - 10:00 Rocket Technology Introduction	The session commenced with an introduction to rocket technology, covering the fundamental principles of space propulsion and launch systems.
10:00 - 10:20 Propulsion System Analysis	A detailed discussion on propulsion mechanisms was conducted, focusing on solid, liquid, and hybrid propulsion systems, with hybrid systems currently being explored by Space Zone India. The session highlighted their working principles, combustion dynamics, thrust generation, and performance efficiency, helping participants understand the advantages and applications of each propulsion type in aerospace engineering.
10:20 - 10:40 Research and Development Insights	Participants were introduced to experimental validation methods, including test videos and combustion variation analysis used to optimize propulsion systems.
10:40 - 11:00 Space Zone India Projects	An overview of ongoing and completed projects was presented: ● RHUMI-H - India's first hybrid rocket ● RHUMI-1 (2024) - featuring movable fins and compact satellite integration ● RHUMI Twin (Update) - next-generation system
11:00 - 11:20 Satellite Technology Overview	The session covered: ● Natural vs Artificial satellites ● Applications of satellites in communication and environmental monitoring ● Doppler effect in satellite communication ● Importance of infrastructure in space research funding.
11:20 - 11:30	REFRESHMENT BREAK

SESSION II: INTERACTION & WEBSITE INAUGURATION 11:30 - 12:20 Interaction with CEO & Website Inauguration	The CEO and Founder of Space Zone India conducted an interactive session with the participants, sharing insights into the space industry, innovation pathways, and future opportunities in aerospace engineering. During this session, the official inauguration of the EWB SVCE student-developed website was carried out. The website was launched by the CEO followed by a brief walkthrough and review. The initiative was appreciated for its technical implementation and relevance, and students were commended for their effort and innovation.
SESSION III: LAB VISITS AND PRACTICAL EXPOSURE 13:00 - 13:50 Aerodynamics Lab	Participants were introduced to fundamental aerodynamic principles, including: • Airflow dynamics in aircraft • Lift and drag forces • Aerodynamics of drones and rockets The session emphasized real-world applications in aviation and UAV systems.
13:50 - 14:40 Rocket Integration Lab	The session focused on the structural and material aspects of rockets, explaining key components such as the nose cone, fins, and body structure. It also covered fuel storage mechanisms and internal configuration, along with materials commonly used in rocket construction, including aluminum and composite polymers. Demonstrations further highlighted practical design principles and integration techniques used in rocket assembly.

14:40 - 15:40 Satellite Lab	Participants explored different types of satellites and their subsystems, including: • Pico satellites • Femto satellites • Cube satellites • Can satellites • Radiation-based satellites
15:40 - 15:50 Valedictory & Photo Session	The visit concluded with a group photo session and informal discussions, marking the successful completion of the program.

CONCLUSION

As the EWB – SVCE Student Chapter, our participation in the **10th All India Chapters Meet (AICM) 2026** at MLR Institute of Technology, Hyderabad, was highly enriching and insightful. Over the two-day national-level convergence, we engaged with student and professional chapters from across India, gaining exposure to innovative community projects, sustainable engineering solutions, and the practical application of Engineering 4.0 skills for societal impact.

The sessions on **Responsible AI and Data-Driven Engineering 4.0**, along with chapter presentations and case studies, provided valuable insights into how engineering interventions can be effectively aligned with the **Sustainable Development Goals (SDGs)**. Interactions with peers, experts, and industry professionals enhanced our understanding of ethical, collaborative, and technically sound approaches to addressing real-world community challenges. Participation in panel discussions and networking activities further strengthened our chapter's perspective on sustainability, leadership, and social responsibility. The meet motivated the SVCE Student Chapter to further innovate, implement impactful projects, and contribute meaningfully to the communities we serve.

Feedback from participants indicated that the event was highly informative and inspiring. Many highlighted that the discussions, project showcases, and collaborative interactions broadened their perspective on socially responsible engineering and encouraged them to actively apply their technical knowledge to community-centered initiatives.

Overall, **AICM 2026** served as a valuable platform for learning, collaboration, and inspiration, reinforcing the collective commitment of EWB chapters to drive sustainable, inclusive, and socially responsible engineering solutions across India.

ANNEXURE – 1

ATTENDEES LIST

FIELD VISIT to SPACE ZONE INDIA, CHENNAI						
PARTICIPANT LIST						
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ANNEXURE 2

GALLERY

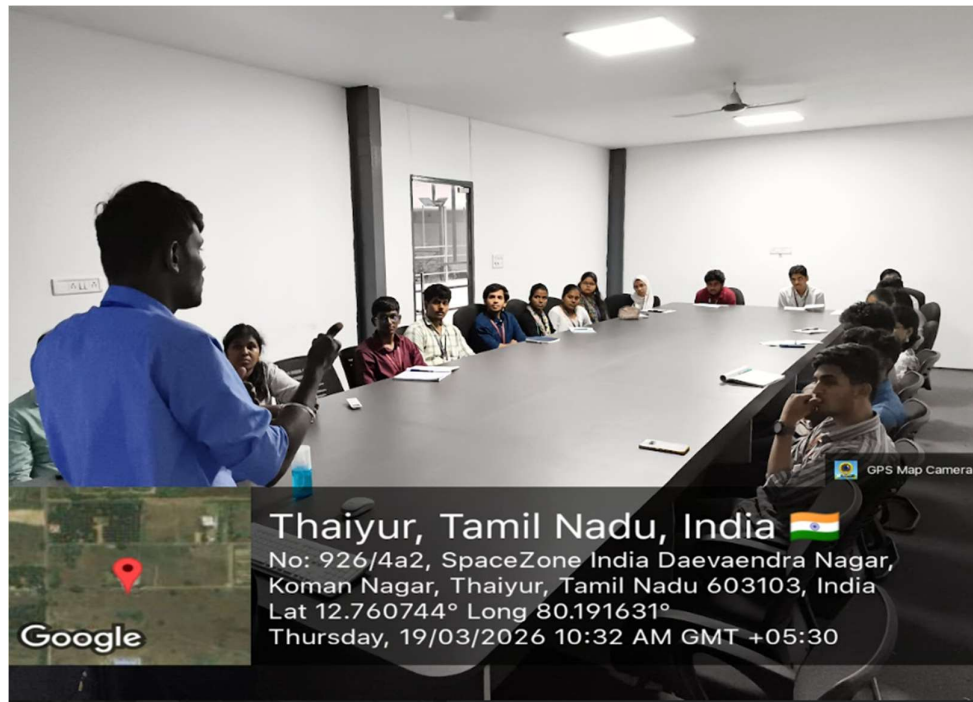


IMAGE - 1 → Technical Session



IMAGE - 2 → Website Inauguration



IMAGE -3 → LABORATORY SESSION



IMAGE - 04 → LABORATORY SESSION



IMAGE - 05 → LABORATORY SESSION



IMAGE - 06 → GROUP PHOTO

ANNEXURE -3

(SNAPSHOTS OF FEEDBACK FORM)

Which part of the visit did you find most interesting?

9 responses

Lab visits

I liked the lab part where each and every process were explained with live materials.

Satellite workspace

Aerodynamics lab

Conversation that i had with the founder and CEO of the space zone india

Rocket technology

Visiting labs

The satellite room where we got to see some satellites launched by the company.

The showcase of all the satellites and the sizes of satellites. Also the part where we were able to view the different components in the rocket felt amusing

Was the session interactive and engaging?

 Copy chart

9 responses



● Yes
● Somewhat
● No

What did you learn from the visit?

9 responses

As i don't even have any basic knowledge about aerospace science and rockets , i learned a lot and it was very informative

I learned how planes and rockets fly. Different types of satellites that are made. How many factors we have to consider for flying and the risk

Yeah we got some basic exposure

I got exposed to rocket technology and I gained knowledge on how my field can contribute to space missions

Got a good idea about basics in rocketry

I learned about how rockets and satellites are designed, tested, and launched. The visit helped me understand the practical aspects of space technology, including propulsion systems, satellite communication, and mission planning. I also gained insight into how organizations like Space Zone India contribute to innovation and hands-on learning in the field of space science. This experience enhanced my interest and gave me a clearer idea about real-world applications in the aerospace domain.

Got information about rockets and satellite.
Got an idea of how ece is applied in rocket science

The information about rockets, satellites and it's launch systems

I learnt the different types of propulsion that were used in rocket launches, and also about satellites and how an object is made to fly

Any additional comments:

5 responses

Good

It was such a wonderful experience and it was very informative and engaging.

Better share the information in a professional manner

Eagerly looking forward to attend more events like this

It Would be better if we had transportation facilities provided by college